

Report of Workshop on Text Encoding Guidelines

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This workshop, held at Vassar College, Poughkeepsie, New York, was funded by the American National Endowment for the Humanities (NEH) with support from Vassar College and organised by the Association for Computers and the Humanities (ACH). The ALLC as well as the Association for Computational Linguistics, was invited to participate at an early stage. The purpose of the workshop was to find some way of defining a consistent scheme for the encoding of textual data for use in humanistic research. To that end a small committee of the ACH had already thrashed out a proposed framework and agreed a list of delegates without whose participation (or at least recognition) no such scheme could stand a chance of survival. Most of the major European text archives were represented (Oxford, Bar-Ilan, Nancy, Oslo, Tübingen, Louvain-la-Neuve, Pisa) as were important North American research centres, including both those purely academic (Provo, Toronto, Pennsylvania) and some bodies with a more commercial affiliation (AAP, BELLCORE, IBM, OCLC). Three observers from the NEH also attended the meeting.

The proceedings began in an atmosphere of gritty resolution against adversity, as the first snowstorm of the 1987–8 winter hit the Eastern seaboard of the United States just as delegates to this historic gathering were making their various ways there. This atmosphere continued as the working hours were long (8 a.m. till 10 p.m. with occasional short breaks for refreshment) and the topics of discussion not trivial (just how do you get 32 different and highly individual delegates to agree on anything?). On the other hand, the organisers (chiefly Nancy Ide of Vassar and Michael Sperberg-McQueen of the University of Illinois at Chicago) had put a great deal of effort into organising the workshop beforehand, and even more into maintaining some structure throughout the event, which together with the evident good will of all participants contributed a great deal to its unusually successful conclusion.

The workshop consisted almost entirely of energetic discussion, the full details of which are beyond the scope of this report. Two delegates remarked on the fact that they had previously attended similar gatherings which had in both cases come to naught. Other than that of the Working Committee's proposed structure for text encoding guidelines by its principal author (Michael Sperberg-McQueen), there were no formal presentations, although several position papers had been circulated before the meeting.

The first session, appropriately enough, attempted to reach a consensus on the scope and nature of the

proposed guidelines. One major area of discussion was whether the guidelines were to be prescriptive—setting out what features should be encoded and how—or descriptive—setting out how those features that had been encoded could be described in a neutral way. A completely open syntax would provide no guidance for those capturing new texts, but a completely prescriptive set of rules would render 90% of existing encoded texts useless. The audience for the guidelines was relevant here: many of those present clearly regarded secondary use of existing machine readable texts, (either from such sources as typesetting tapes or from established archives) as the norm, where others were more concerned about new projects, and the provision of 'guides for the perplexed'. The consensus of this first and wide-ranging discussion was that some kind of 'meta-language' should be defined, capable both of formulating a recommended standard encoding scheme and of describing existing commonly used schemes, almost by way of illustration. It was also agreed that the recommended scheme would be usable as an interchange format, with no implied necessity for retrospective conversions, though it was also pointed out, perhaps somewhat prematurely, that the existence of the Guidelines might be a powerful argument in persuading funding agencies to support such retrospective conversions.

During a very crowded afternoon session, eleven speakers were allowed ten minutes each to describe their own archives and points of view. Stig Johannsson (Oslo) described the work of ICAME, the International Computer Archive of Modern English, an organisation with much experience in documenting and distributing large corpora of machine readable text; Randall Jones (Brigham Young University, Provo) talked about his experience as head of a research support centre with a Kurzweil machine and also a little about the Brigham Young "Wordcruncher" program, Jacques Dendien (INaLF) gave a valuable cautionary tale in the shape of a brief history of the development of the Trésor de la Langue Française database, Paul Tombeur (Louvain) described briefly the CETEDOC Medieval Latin database and stressed the importance of traditional scholarly virtues of fidelity to source and accuracy of description; Robert Kraft (Pennsylvania) talked of the ongoing work of creating both the Thesaurus Linguae Graecae CD-ROM and its Latin counterpart, together with his own concern, ways of representing textual variation in a compact and comprehensible way; Yaacov Choueka (Bar-Ilan) gave a masterly presentation of the problems inherent in handling 100 million words of online Hebrew; I tried briefly to give an idea of the sheer chaos which is the Oxford Text Archive in the absence of any standards; Antonio Zampolli (Pisa) described the evolution of the various corpora of Italian and Latin texts (totalling some 80 million words) at Pisa, all of which are

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encoded and lemmatised to a consistent standard, touching also on the development of the 'linguistic database'; David Barnard (Queens, Ontario) gave a brief introduction to SGML, claiming that it was more than adequate to the tasks so far outlined for a text encoding standard; Frank Tompa (Waterloo) described briefly the Century Dictionary Project, stressing how capturing the layout and typography of such works was usually adequate to capture the underlying structure, provided that the encoding could subsequently be extended and modified; and finally Carol Risher (American Association of Publishers) described the process by which the AAP's SGML Guidelines had been created: this had involved industry-wide co-operation, large-scale funding (the standard had been drafted by an external consultancy and had cost \$450,000 so far) and continued testing, modification and publicity.

After all this, a third discussion session was given over to the question of whether it was meaningful or politically sound to distinguish 'levels' of encoding, as the Working Party had proposed. For some, the use of levels implied a possibly invidious distinction between 'recommended' and 'optional' (where 'recommended' implied 'obligatory for funding support' and 'optional' implied 'not worth the effort'), while for others it implied a possibly unimportant distinction between 'automatically verifiable or capturable' and 'requiring scholarly effort to perceive'. After a straw vote, it was agreed that the guidelines should not make proposals for minimal encoding standards, but rather propose various taggable items under various categories ('boxes'), yet to be defined.

The last session of the first day was held in the library of Alumnae House in a slightly more relaxed atmosphere and concerned itself largely with organisational matters. Several people pointed out the advantages of starting small and getting bigger, (particularly given the fact that some areas of encoding were still a matter of scholarly dispute) while others stressed the need for an overall framework. It was agreed that a small steering committee should meet as soon as possible to set up a committee structure within which more technical discussion could take place and to seek ways of funding this.

After an initial summary of the previous day's proceedings, the second day was taken up almost entirely with discussion of the scope and content of the Guidelines, and the nature of its proposed 'meta-language', concluding with the drafting of a set of recommendations. As a point of departure, Stig Johansson provided a helpful set of statements: the scope of the Guidelines should be 'pieces of extended natural discourse' rather than wordlists, concordances or linguistic surveys. After some discussion, it was agreed that monolingual dictionaries (the major interest of several of those present) should also be covered by the Guidelines. Their purpose should be to facilitate use of such texts in research more widely than by an individual project, rather than to aid conversion of texts to (or from) printed form. Standardisation was needed in three distinct areas: documentation, representation and interpretation. In each case the guidelines should propose what should be included together with indications of how it should be expressed. The end product would not be a formal standard but a

style manual and its production would necessitate working groups in different subject areas.

There was some inconclusive discussion of whether or not SGML provided an appropriate syntax for the definition of the proposed description of encoding schemes; although no-one was able to propose an alternative to the use of SGML, neither were many of the delegates confident enough in their knowledge to criticise (or defend) it, except at a fairly superficial level. There was some consideration of a simplified 'keyboarding' syntax which could be mapped to SGML, though this clearly had little to do with the standard as such. One criticism made of SGML was the difficulty of supporting more than one hierarchy of structural information within a document, though it was claimed that the optional 'concur' feature would support this, another was that the SGML notion of an exhaustive set of document type definitions was fairly inimical to scholarly research, but that these need not be used. On the other hand, several delegates were strongly attracted to the notion of document-type definitions as a means of guiding the perplexed. One telling argument in favour of an SGML-style syntax was its extensibility.

The end product of the meeting was the following text, confected in full session with Michael Sperberg-McQueen at the keyboard, projecting the wordprocessor screen on the wall. Much argument about vocabulary and word order went into its production, but the final product was universally accepted.

The Preparation of Text Encoding Guidelines, Poughkeepsie New York 13 November 1987.

1. The guidelines are intended to provide a standard format for data interchange in humanities research.
2. The guidelines are also intended to suggest principles for the encoding of texts in the same format.
3. The guidelines should
 - a. define a recommended syntax for the format,
 - b. define a metalanguage for the description of text-encoding schemes,
 - c. describe the new format and representative schemes both in that metalanguage and in prose
4. The guidelines should propose sets of coding conventions suited for various applications.
5. The guideline should include a minimal set of conventions for encoding new texts in the format.
6. The guidelines are to be drafted by committees on
 - a. text documentation
 - b. text representation
 - c. text interpretation and analysis
 - d. metalanguage definition and description of existing and proposed schemes, coordinated by a steering committee of representatives of the principal sponsoring organisations
7. Compatibility with existing standards will be maintained as far as possible.
8. A number of large text archives have agreed in principle to support the guidelines in their function as an interchange format. We encourage funding

agencies to support development of tools to facilitate this interchange.

9. Conversion of existing machine-readable texts to the new format involves the translation of their conventions into the syntax of the new format. No requirements will be made for the addition of information not already coded in the texts.

In conclusion, I felt that this workshop had indeed established something, if only in bringing together and actually getting serious discussion out of a very broad-based but surprisingly interdependent constituency of research workers. In the past, there has been almost as much lip-service paid to the notion of a universally-agreed standard for text encoding as there has been to

that of an international directory of machine readable texts; with this workshop, for the first time, a significant proportion of those without whose support all such notions must founder had actually devoted their undivided attention to the issue and the problems it raises for at least two days. The organisers (and participants!) deserve every credit for this achievement, still more for having orchestrated a productive kind of consensus from which something more tangible may well eventually emerge.

[A note from the ALLC Chairman on the planning of further stages of this project appears in the News and Notes section of this issue]